Arch Orthop Trauma Surg
DOI 10.1007/s00402-016-2610-x

ORTHOPAEDIC SURGERY

The burden of rotator cuff surgery in Italy: a nationwide registry study

Umile Giuseppe Longo[1] · Giuseppe Salvatore[1] · Giacomo Rizzello[1] · Alessandra Berton[1] · Mauro Ciuffreda[1] · Vincenzo Candela[1] · Vincenzo Denaro[1]

Received: 20 September 2016
© Springer-Verlag Berlin Heidelberg 2016

Abstract
Purpose This study aimed to estimate the yearly number of RC surgeries in Italy, a country with universal access to healthcare for its population, from 2001 to 2014, based on official information source as hospitalization records. A secondary aim was to explore geographical variation in equity in access to RC surgery between three macroregions of Italy (North, Center and South). A tertiary aim was to perform statistical projections of the number of RC procedure volumes and rates on the basis of data from 2001 to 2014. Finally, we aimed to perform a prediction of charges by 2025 to examine the economic impact of RC surgery.
Methods The analysis of the National Hospital Discharge records (SDO) maintained at the Italian Ministry of Health, concerning the 14 years of our survey (2001 through 2014) was performed. These data are anonymous and include the patient's age (in aggregate for class of age), sex, domicile, region of hospitalization, length of the hospitalization, and type of reimbursement (public or private).
Results During the 14-year study period, 390,001 RC repairs were performed in Italy, which represented an incidence of 62.1 RC procedures for every 100,000 Italian inhabitants over 25 years old. Approximately 65% of RC repair were performed annually in patients ages <65 years, thus affecting the working population. 246,810 patients (63.3%) from the North underwent RC repairs from 2001 through 2014, 78,540 patients (20.2%) from the Center, and 64,407 patients (16.5%) from the South. The projection model predicted substantial increases in the numbers of RC repairs.
Conclusions This study confirms that the socioeconomic burden of RC surgery is growing and heavily affecting the working population. According to the prediction model, hospital costs sustained by the national health care system for RC procedures are expected to be over 1 billion euros by 2025.
Level of evidence II.

Keywords Burden · Rotator · Cuff · Surgery · Nationwide · Registry

Introduction

Rotator cuff (RC) disease is frequent and represents a common source of shoulder pain [4, 8]. Each year in the U.S. approximately 4.5 million patient visits related to shoulder pain occur, and the majority of them are due to RC problems [10]. Full-thickness RC tears may occur in roughly 20% of the general population [24].

Patients with RC pathology may complain of symptoms ranging from minimal discomfort without functional deficits to severe pain, weakness, and marked disability [15]. Surgical repair of RC tears is a well-documented therapeutic option in these patients [5, 14].

Geographic variation in the number of RC surgical procedures performed in the US and variations in what surgeons deemed to be indications for RC surgery has been studied [16, 20]. Several reports have highlighted the apparent broadening of indications for RC surgery [3, 17, 20, 23], showing a considerable variation in the treatment of RC disease and the conditions requiring surgical repair. Incidence of RC surgery is rapidly increasing

✉ Umile Giuseppe Longo
ug.longo@gmail.com

[1] Department of Trauma and Orthopaedic Surgery, University Campus Bio-Medico of Rome, Via Álvaro Del Portillo, 200, 00128 Rome, Italy

Published online: 08 December 2016

in selected patient cohorts [6], but nationwide incidence rates have been reported only for Finland [18].

Despite several publications address RC biology [2, 9, 13], indications for surgery, and outcomes in selected groups of patients [11, 21, 22, 24], little information is available on trends of use.

Understanding of the trends of variability in the RC repair rates could be important to better address the future evolution of this procedure and the related health service planning [24, 25].

One of the founding principles of the Italian National Health Service (NHS) is fairness in access to health care. The Italian NHS is free to patients at the point of use.

This study aimed to estimate the yearly number of RC surgeries in Italy, a country with universal access to healthcare for its population, from 2001 to 2014, based on official information source as hospitalization records. A secondary aim was to explore geographical variation in equity in access to RC surgery between three macroregions of Italy (North, Center and South). A tertiary aim was to perform statistical projections of the number of RC procedure volumes and rates on the basis of data from 2001 to 2014 by 2015. Finally, we aimed to perform a prediction of charges by 2025 to examine the economic impact of RC surgery.

Materials and methods

The analysis of the National Hospital Discharge records (SDO) maintained at the Italian Ministry of Health, concerning the 14 years of our survey (2001 through 2014) was performed. In this archive, information concerning all hospitalizations occurring in Italian public and private care settings is collected. These data are anonymous and include the patient's age (in aggregate for class of age), sex, domicile, region of hospitalization, length of the hospitalization, and type of reimbursement (public or private).

Population data were obtained from the National Institute for Statistics (ISTAT) for each year.

RC repair was defined by the following International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM) major diagnosis code: 83.63 (Rotator Cuff repair).

The operative treatment code 83.63 (rotator cuff repair) is used for both open and arthroscopic RC repair. Therefore, this coding system did not allow us to differentiate between open and arthroscopic approaches used for RC repairs performed during the study period.

Several procedures can be performed during same RC operation. We wanted also to examine concomitant procedures associated with RC repair (such as acromioplasty, tenotomy of the long head of the biceps tendon, tenodesis of the long head of the biceps tendon, and resection of the distal end of the clavicle), but unfortunately these procedures do not have a specific code, and often are not codified in the SDO.

Since the total number of RC surgeries for the class of age 0–25 years was less than 1500 over the 14-year study period, to avoid underestimation of the population which may suffer from RC disease, we decided to refer the study to the Italian population over 25 year of age.

Italian geographical macroregions are the North, Center and South (Fig. 1).

The North includes the regions of the North–West (Liguria, Lombardy, Piedmont, Aosta Valley) and those of the North–East (Emilia–Romagna, Friuli–Venezia Giulia, Trentino—South Tyrol, Veneto). The Center includes the regions of Lazio, Marche, Tuscany, Umbria.

The South includes the regions of Southern Italy (Abruzzo, Basilicata, Calabria, Campania, Molise, Apulia) and the islands (Sardinia and Sicily).

DRG

Analyses of estimated costs were based on the costs ascribed to diagnosis-related groups (DRGs), according to Ministerial Decree (December 18, 2008).

In Italy reimbursement is the same for all the procedures under diagnosis-related group (DRG), regardless of the

Fig. 1 Italian geographical macroregions are the North, Center and South

diagnosis, the complexity of the procedure, or the patient's health status at admission.

Costs for rehabilitation are not evaluable, because in Italy all patients undergoing RC repair are treated on an outpatient basis.

Statistics

Descriptive statistical analyses were used to calculate the annual number of RC repair in the whole Italian population. Incidence rates were calculated using the annual adult population size obtained from ISTAT, a statutory electronic national population register. The incidence was based on the size of the entire population of people ≥25 year old in Italy.

Projection was performed using the "Forecast" function in Excel (Microsoft) software.

Results

Demographics

During the 14-year study period, 390,001 RC repairs were performed in Italy (Fig. 2), which represented an incidence of 62.1 RC procedures for every 100,000 Italian inhabitants over 25 years old (Table 1).

From 2001 to 2014, the incidence of operations increased from 28.6 to 73.5 per 100,000 person-years over 25 years old, with a peak in 2012 (77.6 per 100,000 person-years over 25 years old).

Over the study period, the highest incidence of surgeries was found in the 45- to 64-year age group (Fig. 3).

Approximately 65% of RC repair were performed annually in patients ages <65 years, thus affecting the working population.

The number of RC surgeries in Italy during the study period by gender is reported in Fig. 4. Men represented the slight majority of patients undergoing RC procedures (average male:female ratio 1.02:1). From 2001 to 2014, the male to female ratio increased from 0.94:1 in 2001 to 1.1:1 in 2014.

Length of the hospitalization

The median length of hospital stay was 2 days (range 0–4 days). The median length of hospital stay for each year is reported in Table 1.

Region of hospitalization and domicile of the patients

Regarding regional distribution of surgeries, there were 265,501 cases of RC repairs identified in the North (68.1%) from 2001 through 2014, whereas the Center had 74,563 cases (19.1%), and the South 49,937 cases (12.8%) (Fig. 5).

Of 390,001 RC repairs performed in Italy during the study period, data on domicile of the patient was not available for 244 RC repairs, thus resulting in a total of 389,757 RC repairs for which data on domicile of the patients was available.

246,810 patients (63.3%) from the North underwent RC repairs from 2001 through 2014, 78,540 patients (20.2%) from the Center, and 64,407 patients (16.5%) from the South (Fig. 6).

Considering the Italian population of the three different regions of Italy (North, Center and South), from 2001 to 2014, the incidence of operations was 77.7 per 100,000 person-years in the North, 35.5 per 100,000 person-years in the Center, 22.2 per 100,000 person-years the South.

Public or private surgery

From 2001 to 2014, the incidence rate of private RC repair was 11,723 (3.04%).

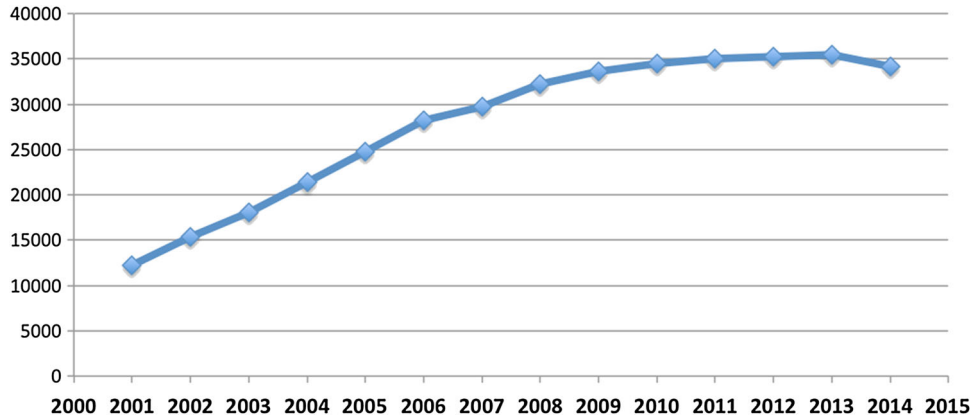


Fig. 2 Number of RC repair for each year during the study period

Table 1 Data on demographics, length of the hospitalization, region of hospitalization, domicile of the patients, public or private surgery

	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
National Population >25 year old (ISTAT Database)	42,538,938	42,987,625	43,566,254	44,107,797	44,395,796	44,758,401	45,171,957	45,417,194	45,776,638	45,933,926	45,149,939	45,425,592	46,357,295	46,465,918
Total of RC repairs	12,178	15,402	18,046	21,415	24,778	28,203	29,772	32,274	33,616	34,476	34,991	35,229	35,479	34,140
Male	5897	7345	8663	10,246	12,205	13,895	14,727	16,081	17,145	17,685	18,172	18,388	18,574	18,019
Female	6281	8057	9383	11,169	12,573	14,308	15,045	16,193	16,471	16,791	16,819	16,841	16,905	16,121
Male/female ratio	0.94	0.91	0.92	0.92	0.97	0.97	0.98	0.99	1.04	1.05	1.08	1.09	1.10	1.12
Region of hospitalization														
North	9022	11,244	12,618	14,804	16,876	19,072	20,547	22,283	22,802	23,230	23,451	23,407	23,731	22,414
Center	2181	2773	3520	4241	5024	5514	5531	5854	6497	6602	6583	6772	6701	6770
South	975	1385	1909	2370	2878	3618	3694	4137	4317	4644	4957	5050	5047	4956
No. of RC repairs per 100.000 inhabitants	28.63	35.83	41.42	48.55	55.81	63.01	65.91	71.06	73.43	75.06	77.50	77.55	76.53	73.47
North	40.24	49.66	54.36	62.29	70.66	79.63	83.95	90.14	90.88	91.52	94.13	93.16	92.98	87.91
Center	15.04	18.72	23.74	28.50	33.34	35.03	35.75	37.98	42.68	43.58	45.77	45.42	43.39	43.12
South	6.05	8.66	11.87	14.98	17.85	22.42	23.40	25.66	26.74	28.54	29.92	31.47	31.49	30.83
Public surgery per 100.000 inhabitants	20.7	26.2	30.3	35.3	40.7	46.2	48.4	48.1	51.8	55.2	57.1	57.5	56.7	54.7
Private surgery per 100.000 inhabitants	0.7	0.7	0.9	1.4	1.5	1.5	1.6	1.5	1.7	1.7	1.8	1.5	1.7	1.5
Length of hospitalization (days)	3.9	3.3	3.0	2.8	2.6	2.5	2.3	2.2	2.1	1.9	1.9	1.8	1.8	1.7

The incidence rate of RC repairs in public hospitals increased from 20.7 to 54.7 per 100,000 person-years.

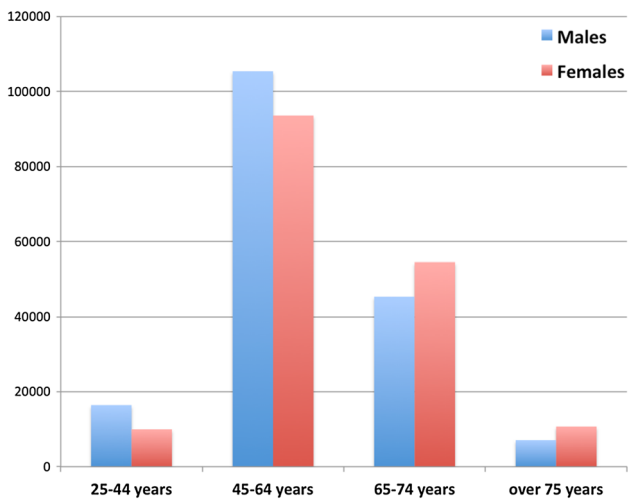


Fig. 3 Distribution of total RC repair by age group and by gender

The incidence in private hospitals increased during the study period, from 0.7 to 1.5 operations per 100,000 person-years.

In 2001, 96.77% of RC repairs were performed in public hospitals and 3.23% in private hospitals. In 2014, the corresponding percentages were 97.33 and 2.67%.

Projection

The projection model predicted substantial increases in the numbers of RC repairs (Fig. 7). On the basis of the projection model, the demand for RC repairs was estimated to grow by 172.66%, from 34,140 in 2014 to 57,375 by 2025.

Estimated economic impact

The actual mean hospital reimbursement is 2125.28€ for each RC procedure. According to the prediction model, hospital costs sustained by the national health care system for RC procedures are expected to be over 1 billion euros (1,051,982,935.25) by 2025.

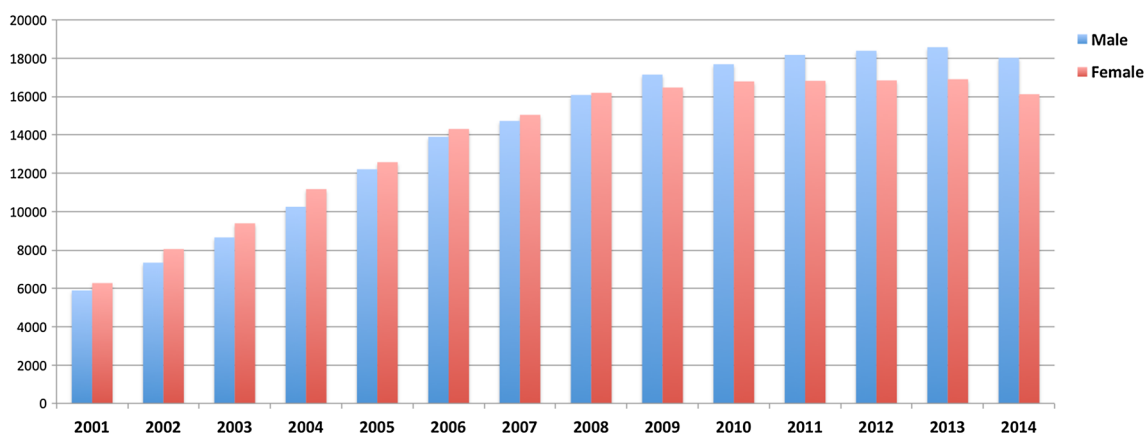
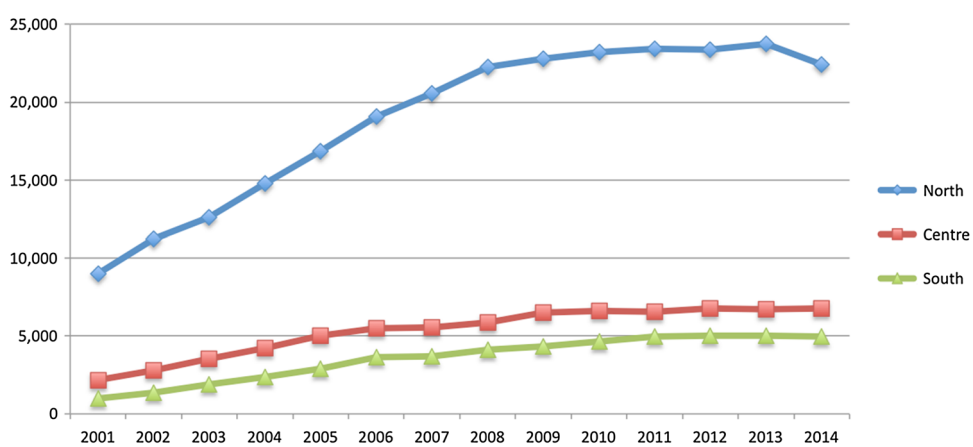


Fig. 4 Male to female ratio during the study period. *Each bar* represents the number of patients undergoing RC repair, stratified by sex (*blue* for male and *red* for female)

Fig. 5 Regional distribution of surgeries for each year during the study period



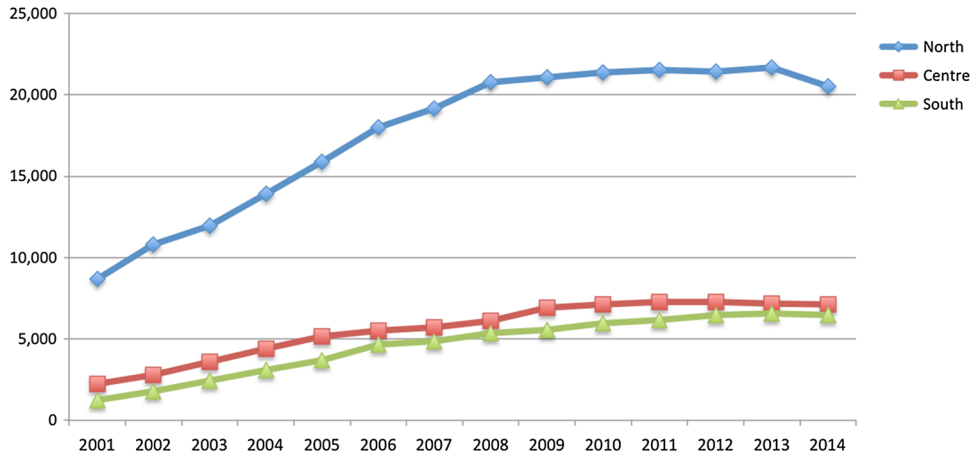


Fig. 6 Domicile of patients at the time of surgery for each year during the study period

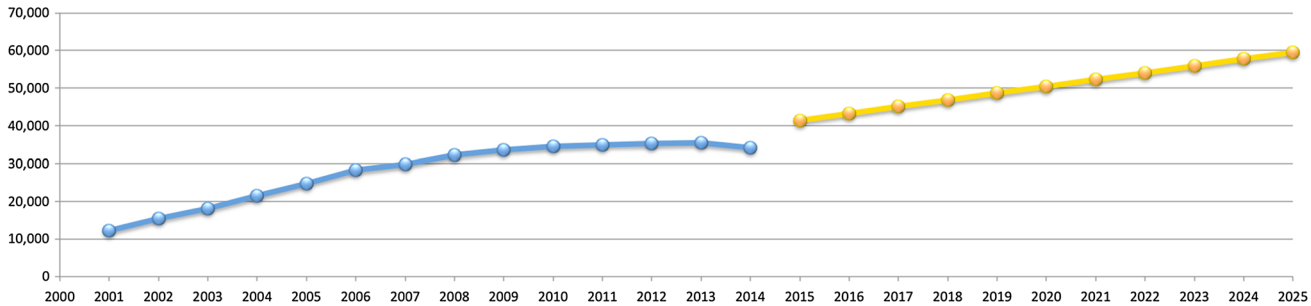


Fig. 7 The projection model predicted substantial increases in the numbers of RC repairs

Discussion

The most important results of this study are that rate of RC repair in Italian hospitals is increased between 2001 and 2014, confirming that the socioeconomic burden of RC surgery is growing. Approximately 65% of RC repair were performed annually in patients ages <65 years, thus heavily affecting the working population. Over a 14-year study time, geographical variations in RC repair rates were present among Italian macroregions. The majority of RC procedures were performed in the northern part of Italy. According to the prediction model, hospital costs sustained by the national health care system for RC procedures are expected to be over 1 billion euros by 2025.

It is not possible to discriminate whether these differences reflect variations in need or in clinical practice, because data on prevalence, incidence, and severity of RC tears, and information regarding clinicians' and patients' attitudes to RC surgery are not available [19]. They may also indicate differing amounts of resources available to provide RC surgery.

Growing awareness of RC disorders, increased availability of diagnostic methods and better knowledge of arthroscopy may be the potential explanations for this increasing trend.

The findings of this study with regards to regionalization are interesting. In particular, we found differences in the Italian population of the three different macroregions of Italy (North, Center and South), with incidence of operations of 77.7 per 100,000 person-years in the North, 35.5 per 100,000 person-years in the Center, and 22.2 per 100,000 person-years the South. In view of the government and Italian Ministry of Health focus on reducing health inequalities, the difference in RC surgeries among regions of Italy highlights an area that warrants further investigation. It is possible that these differences reflect real geographic variation in need or patient consent to surgery. However, some studies have suggested that need is greater among the most deprived groups, and that education and income are unrelated to willingness to undergo surgery [1]. If so, this would imply inequity in the provision of RC surgery. A nationwide registry study in Finland showed higher incidence rate (from 44 to 131 per 100,000 adults per year) compared to rates in Italy [18].

The effects of non-surgical treatment of RC tear are currently unclear. A recent randomized controlled trial compared physiotherapy; acromioplasty and physiotherapy; and RC repair, acromioplasty, and physiotherapy in the treatment of symptomatic, nontraumatic, RC tears among patients older than fifty-five years of age with two

years of follow-up [7]. The authors found no significant difference in clinical outcome between the three interventions at the two-year follow-up. This is the only level I evidence study in the literature to support conservative treatment for RC cuff tears. Unfortunately, the size of the tear increased slightly in the non-repaired groups [7]. Degenerative findings, including osteoarthritis and muscular fatty infiltration as seen on MRI, demonstrated global tissue degeneration of the studied shoulders [7]. The possible progression of RC tears, especially in the non-repaired treatment groups, warrants further follow-up and it can be of concern, especially when treating younger patients [12]. Increased waiting time for surgery could be responsible for fatty infiltration, narrowing of the acromiohumeral distance and increasing of the size of the lesion. Small- and medium-sized tears, if left surgically untreated, can become large and massive, with associated muscle atrophy and fatty infiltration within the RC muscles.

The main concern in clinical practice is that "reparable" tears could become "irreparable" during the period of waiting list [12]. Fatty degeneration is irreversible even with repair and leads to reduced function of the RC musculature.

Few studies have evaluated the incidence of surgical treatment across age groups. In one study from the US, the highest incidence of RC repair was found in the 50- to 59-year age group, followed closely by the 60- to 69-year age group, and then the 40- to 49-year age group [26]. The results of this study are in accord to these results.

Our finding that hospital length of stay decreased over the 14-year study period is somewhat expected, given the fact that hospitals, under intense financial pressure, have reduced hospital length of stay as much as possible, resulting in this difference.

The total amount of private procedures remained constant (about 3%) over the 14-year study period. The main reason for this is that the one of the founding principles of the Italian National Health Service (NHS) is fairness in access to health care. The Italian NHS is free to patients at the point of use.

Strengths of this study included its national coverage, long time span, the use of reliable nationwide data and that the study population comprised the entire adult population of Italy. Data collection for the Italian Ministry of Health is mandatory for all hospitals, including public and private institutions.

Since the data cover the entire country, including both public and private hospitals, the observed changes in RC repair incidence are representative of nationwide clinical practice and trends for all actively practicing orthopedic surgeons in Italy.

One limitation of this study could be that it relied upon administrative data, and thus there is the possibility that changes in coding practices could have impacted our results. That said we are unaware of any systematic changes in the coding of RC repair which would have significantly impacted our findings. Significant regional variations were found in our study, confirming an inequity. This may express unmet needs in the low RC repair rate areas, or else over-indication in the high RC repair rate areas. So, part of this variability may be related to patients and to surgeons, and this cannot be clarified from the database analysis.

A limitation of this registry study is that we are unable to evaluate potential inaccuracies in coding of the diagnoses or procedures. Another limitation of our study is that the diagnosis and procedure coding used during the study period did not allow differentiation between open and arthroscopic procedures, or between partial and full-thickness tears.

Conclusion

In conclusion, this study confirms that the socioeconomic burden of RC surgery is growing and heavily affecting the working population. According to the prediction model, hospital costs sustained by the national health care system for RC procedures are expected to be over 1 billion euros by 2025. We found evidence of a concentration of RC procedures in the north of Italy. Our results give pause over whether efforts on regionalization of RC repair should turn towards improving quality in hospitals in the South of Italy. There is an evidence of inequity in access to RC surgeries across macroregions of Italy. Today's levels of RC surgery are below the expected maximum incidence, and we expect a continued annual increase in the total number of RC repair performed. Policy makers should examine factors to understand the determinants of inequitable provision.

Acknowledgements We thank the Direzione Generale della Programmazione Sanitaria—Banca Dati SDO of the Italian Ministry of Health for the support in providing data for this research.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Funding There is no funding source.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

References

1. Chaturvedi N, Ben-Shlomo Y (1995) From the surgery to the surgeon: does deprivation influence consultation and operation rates? Br J Gen Pract 45(392):127–131

2. Denaro V, Ruzzini L, Longo UG, Franceschi F, De Paola B, Cittadini A, Maffulli N, Sgambato A (2010) Effect of dihydrotestosterone on cultured human tenocytes from intact supraspinatus tendon. Knee Surg Sports Traumatol Arthrosc 18(7):971–976. doi:10.1007/s00167-009-0953-3
3. Dunn WR, Schackman BR, Walsh C, Lyman S, Jones EC, Warren RF, Marx RG (2005) Variation in orthopaedic surgeons' perceptions about the indications for rotator cuff surgery. J Bone Joint Surg Am 87(9):1978–1984. doi:10.2106/JBJS.D.02944
4. Ensor KL, Kwon YW, Dibeneditto MR, Zuckerman JD, Rokito AS (2013) The rising incidence of rotator cuff repairs. J Shoulder Elbow Surg 22(12):1628–1632. doi:10.1016/j.jse.2013.01.006
5. Franceschi F, Longo UG, Ruzzini L, Rizzello G, Maffulli N, Denaro V (2008) Soft tissue tenodesis of the long head of the biceps tendon associated to the Roman Bridge repair. BMC musculoskeletal disorders 9:78. doi:10.1186/1471-2474-9-78
6. Judge A, Murphy RJ, Maxwell R, Arden NK, Carr AJ (2014) Temporal trends and geographical variation in the use of subacromial decompression and rotator cuff repair of the shoulder in England. Bone Joint J 96-B(1):70–74. doi:10.1302/0301-620X.96B1.32556
7. Kukkonen J, Joukainen A, Lehtinen J, Mattila KT, Tuominen EK, Kauko T, Äärimaa V (2015) Treatment of Nontraumatic Rotator Cuff Tears: A Randomized Controlled Trial with Two Years of Clinical and Imaging Follow-up. J Bone Joint Surg Am 97(21):1729–1737. doi:10.2106/JBJS.N.01051
8. Longo UG, Berton A, Ahrens PM, Maffulli N, Denaro V (2011) Clinical tests for the diagnosis of rotator cuff disease. Sports Med Arthrosc 19(3):266–278. doi:10.1097/JSA.0b013e3182250c8b
9. Longo UG, Berton A, Khan WS, Maffulli N, Denaro V (2011) Histopathology of rotator cuff tears. Sports Med Arthrosc 19(3):227–236. doi:10.1097/JSA.0b013e318213bccb
10. Longo UG, Berton A, Papapietro N, Maffulli N, Denaro V (2012) Epidemiology, genetics and biological factors of rotator cuff tears. Med Sport Sci 57:1–9. doi:10.1159/000328868
11. Longo UG, Franceschi F, Berton A, Maffulli N, Denaro V (2012) Arthroscopic transosseous rotator cuff repair. Med Sport Sci 57:142–152. doi:10.1159/000328900
12. Longo UG, Franceschi F, Berton A, Maffulli N, Droena V (2012) Conservative treatment and rotator cuff tear progression. Med Sport Sci 57:90–99. doi:10.1159/000328910
13. Longo UG, Franceschi F, Ruzzini L, Rabitti C, Morini S, Maffulli N, Denaro V (2008) Histopathology of the supraspinatus tendon in rotator cuff tears. Am J Sports Med 36(3):533–538. doi:10.1177/0363546507308549
14. Longo UG, Franceschi F, Spiezia F, Marinozzi A, Maffulli N, Denaro V (2011) The low-profile Roman bridge technique for knotless double-row repair of the rotator cuff. Arch Orthop Trauma Surg 131(3):357–361. doi:10.1007/s00402-010-1203-3
15. Longo UG, Vasta S, Maffulli N, Denaro V (2011) Scoring systems for the functional assessment of patients with rotator cuff pathology. Sports medicine and arthroscopy review 19(3):310–320. doi:10.1097/JSA.0b013e31820af9b6
16. Mall NA, Lee AS, Chahal J, Sherman SL, Romeo AA, Verma NN, Cole BJ (2013) An evidenced-based examination of the epidemiology and outcomes of traumatic rotator cuff tears. Arthroscopy 29(2):366–376. doi:10.1016/j.arthro.2012.06.024
17. Mauro CS, Jordan SS, Irrgang JJ, Harner CD (2012) Practice patterns for subacromial decompression and rotator cuff repair: an analysis of the American Board of Orthopaedic Surgery database. J Bone Joint Surg Am 94(16):1492–1499. doi:10.2106/JBJS.J.01696
18. Paloneva J, Lepola V, Äärimaa V, Joukainen A, Ylinen J, Mattila VM (2015) Increasing incidence of rotator cuff repairs–A nationwide registry study in Finland. BMC Musculoskelet Disord 16:189. doi:10.1186/s12891-015-0639-6
19. Randelli P, Arrigoni P, Cabitza F, Ragone V, Cabitza P (2012) Current practice in shoulder pathology: results of a web-based survey among a community of 1,084 orthopedic surgeons. Knee Surg Sports Traumatol Arthrosc 20(5):803–815. doi:10.1007/s00167-011-1673-z
20. Vitale MG, Krant JJ, Gelijns AC, Heitjan DF, Arons RR, Bigliani LU, Flatow EL (1999) Geographic variations in the rates of operative procedures involving the shoulder, including total shoulder replacement, humeral head replacement, and rotator cuff repair. J Bone Joint Surg Am 81(6):763–772
21. Werner BC, Brockmeier SF, Gwathmey FW (2015) Trends in long head biceps tenodesis. Am J Sports Med 43(3):570–578. doi:10.1177/0363546514560155
22. White JJ, Titchener AG, Fakis A, Tambe AA, Hubbard RB, Clark DI (2014) An epidemiological study of rotator cuff pathology using The Health Improvement Network database. Bone Joint J 96-B(3):350–353. doi:10.1302/0301-620X.96B3.32336
23. Wolf BR, Dunn WR, Wright RW (2007) Indications for repair of full-thickness rotator cuff tears. Am J Sports Med 35(6):1007–1016. doi:10.1177/0363546506295079
24. Yamamoto A, Takagishi K, Osawa T, Yanagawa T, Nakajima D, Shitara H, Kobayashi T (2010) Prevalence and risk factors of a rotator cuff tear in the general population. J Bone Joint Surg Am 19(1):116–120. doi:10.1016/j.jse.2009.04.006
25. Yu E, Cil A, Harmsen WS, Schleck C, Sperling JW, Cofield RH (2010) Arthroscopy and the dramatic increase in frequency of anterior acromioplasty from 1980 to 2005: an epidemiologic study. Arthroscopy 26(9 Suppl):S142–S147. doi:10.1016/j.arthro.2010.02.029
26. Zhang AL, Montgomery SR, Ngo SS, Hame SL, Wang JC, Gamradt SC (2013) Analysis of rotator cuff repair trends in a large private insurance population. Arthroscopy 29(4):623–629. doi:10.1016/j.arthro.2012.11.004